

PSCAD Build Plan: Double-Conversion UPS-Microgrid with PCS3440

Detailed Modeling Tasks for AI Factory Load Swing Mitigation

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*Key Specs: Load step 20→90% within 16 ms (1 cycle), mode transition ≤ 4 ms (¼ cycle),
input ramp ≤ 20 MW/min, intra-minute ≤ 10 MW/5 s*

1. Objective and System Context

This document provides a step-by-step PSCAD/EMTDC build plan for a double-conversion UPS-microgrid based on the Delta PCS3440 platform serving large AI factory loads. The simulation must reproduce the three operating modes (Double-Conversion, Bypass, Backup) and validate that the system meets the following hard requirements driven by AI training/inference load behavior and TSO/TCLL (e.g., Southern Company TCLL v1.2) interconnection rules.

Requirement	Target
Output V/f deviation under 20→90% load step	≤ 5% within 16 ms (1 cycle)
Mode transition time	≤ 4 ms (¼ cycle)
Input ramp rate to utility	≤ 20 MW/min
Intra-minute variation at PCC	≤ 10 MW per 5 s window
Reactive support	STATCOM in Bypass mode, ±0.95 PF
Block field size	5 × 28 MW = 140 MW total

2. PSCAD Project Structure

Build a hierarchical workspace so that one 28 MW block can be developed, validated, then duplicated 5× without rework. Switching-level detail is only kept inside the PCS3440 page module; the rest of the plant uses averaged equivalents to keep simulation time reasonable.

- Top page (PLANT): 34.5 kV utility source + gas generator + Breakout breaker + 5× block feeders + Aggregation bus + 3000 A STS + AI load model.

- Page module BLOCK_28MW: 1× Power Conversion Skid (PCS3440 input + output back-to-back + XFOs) + Battery DC block + local UPS-Microgrid Controller.
- Page module PCS3440_UC: Unit controller (input converter + output converter + DC link manager + mode arbiter).
- Page module UPS_MGC: Block-level coordination, mode state machine, STS command, ramp shaper.
- Page module PLANT_SCADA: Plant-level dispatch, EMS interface, 5-block aggregation.
- Page module AI_LOAD: Programmable load profile generator (steps, ramps, training/inference patterns).
- Time step: $\Delta t = 20 \mu s$ main, $50 \mu s$ for averaged blocks, controller multi-rate $\times 5$.

3. Electrical Model Build Tasks

3.1 Source and Backbone

- Task E1 — Create a 34.5 kV Thevenin source (configurable SCR = 5–20) representing utility + gas generator parallel feed.
- Task E2 — Add 34.5 kV breakout breaker with controllable open/close from PLANT_SCADA.
- Task E3 — Build five 600 A feeders, each rated for 28 MW @ 34.5 kV; model as π -section cables (~ 100 m typical).
- Task E4 — Aggregation bus and 3000 A static transfer switch (STS) modeled as two anti-parallel thyristor pairs with gate logic.

3.2 PCS3440 Power Conversion Skid

- Task E5 — Step-up XFO (input side): 34.5 kV / 690 V Δ -Yg, leakage 6%, rated 4.3 MVA per string PCS.
- Task E6 — Input converter: 3-phase 2-level VSC with LCL filter ($L_f=0.05$ pu, $C_f=0.07$ pu, $L_g=0.03$ pu).
- Task E7 — DC link: C_{dc} sized for 8–10 ms ride-through at full power ($C_{dc} \approx 2 \cdot P \cdot \Delta t / (V_{dc}^2 - V_{min}^2)$).
- Task E8 — Output converter: identical VSC topology, supplies regulated 480 V or 690 V to the AI load via output XFO.
- Task E9 — Output XFO: 690 V / 480 V Yg-Yg (or 34.5 kV/690 V if back-to-the-medium-voltage), 6% Z.
- Task E10 — Battery DC block: Thevenin/RC battery model (1000–1500 V_{dc} nominal), with SoC integrator and current-limit clamp.

- Task E11 — Replicate Task E5–E10 to form 4–7 parallel PCS groups per 28 MW block (use PSCAD subpage instances).

3.3 AI Load Model

AI factories produce highly dynamic active power profiles — large step changes from idle to full training load, deep dips during checkpoint/inference, and sub-second pulsation. The PSCAD load must allow programmable patterns.

- Task E12 — Variable resistive + constant-power composite load (CPL) up to 140 MW, with controllable P_{ref} input.
- Task E13 — Profile generator with five preset patterns: (a) 20%→90% step, (b) 90%→20% step, (c) training rhythm (1 Hz pulsation), (d) inference (constant), (e) checkpoint dip.
- Task E14 — Load also draws Q proportional to P ($PF=0.95$ – 1.0 selectable).
- Task E15 — File-input mode: read .txt profile (time, P , Q) from disk to replay real AI factory traces.

4. PCS3440 Unit Controller — Detailed Build Tasks

4.1 Input-Side Converter Controller

- Task C1 — Voltage/current sensing: V_{abc} , i_{abc} at LV side of input XFO; V_{dc} at DC link.
- Task C2 — SRF-PLL on utility-side V_{abc} , bandwidth 20–40 Hz, lock to grid θ_{grid} .
- Task C3 — $abc \rightarrow dq$ using θ_{grid} ; produce I_d , I_q , V_d , V_q feedback.
- Task C4 — Outer DC-link voltage loop: $V_{dc_ref} = 1200$ V (nominal). PI: $K_p=2$, $K_i=120$ (start values), bandwidth 30–80 Hz.
- Task C5 — RAMP SHAPER (key for AI mitigation): place a rate-limited filter on I_d^* with $slew = 20$ MW/min/ N_{pcs} converted to A/s. This forces the utility-side current to ramp slowly even if V_{dc} drops fast.
- Task C6 — Inner dq current loop: bandwidth 600–1000 Hz, cross-decoupling ($\pm \omega L \cdot I_q$, $\mp \omega L \cdot I_d$), feed-forward V_d/V_q .
- Task C7 — Modulation: SVPWM, carrier 2.5 kHz, dead-time 3 μ s.
- Task C8 — Reactive support: I_q^* generated by a Q-control loop or grid voltage support (Q-V droop 4%).
- Task C9 — Protection: dq over-current trip (>1.5 pu, 1 ms), DC OV/UV (1300/950 V), grid OV/UV per IEEE 1547/2800.

4.2 Output-Side Converter Controller

- Task C10 — Output PLL: synthesize internal θ_{out} at 60 Hz (grid-forming).

- Task C11 — Outer voltage loop in dq: $V_d\text{_ref}$ = nominal, $V_q\text{_ref}$ = 0. PI bandwidth 200–400 Hz so a 20→90% load step is corrected within ≤ 16 ms with $<5\%$ sag.
- Task C12 — Inner current loop: bandwidth 1.5–2 kHz; this fast loop is what delivers the 1-cycle response.
- Task C13 — Feed-forward of load current ($i\text{_load}$) into V_d loop output → eliminates 5–8 ms of inherent PI lag during a step.
- Task C14 — Virtual impedance / droop block for paralleling multiple output converters in one block (4–7 PCS share kVAR).
- Task C15 — Voltage harmonic compensator (5th, 7th, 11th, 13th) to keep THD $< 3\%$ under nonlinear AI load.
- Task C16 — Output protection: OC limiter, soft-start ramp on $V_d\text{_ref}$ at black-start.

4.3 DC-Link / Battery Manager

- Task C17 — Power balance equation: $P\text{_dc} = P\text{_in} - P\text{_out} - P\text{_loss}$; integrate to track Cdc state.
- Task C18 — Battery current command $I\text{_batt}^* = (P\text{_out_smooth} - P\text{_in_smooth}) / V\text{batt}$; this is the heart of AI swing mitigation — battery absorbs the fast Δ between load and ramp-limited grid draw.
- Task C19 — SoC limits (10–90%), current de-rating curves, thermal model (1st-order).
- Task C20 — Battery dq/dt limiter only as protective fallback; primary smoothing comes from the input ramp shaper (Task C5).

4.4 Mode State Machine inside UC

State	Input Conv	Output Conv	STS	Battery
Double-Conversion	GFL, Vdc reg, ramp-limited	GFM, V/f reg	Open	Active (charge/discharge)
Bypass	STATCOM (Q only)	STATCOM or Off	Closed	Float charge
Backup	Disconnected	GFM, V/f reg	Open	Discharging
DC→Bypass transition	Reduce P linearly	Reduce P linearly	Close at ZCD	Hold
Bypass→Backup (unplanned)	Trip	Pre-armed, takeover	Open at fault detect	Discharge

- Task C21 — Implement state machine in PSCAD using ‘Multiple Run’ + custom logic component or external C/Fortran routine.
- Task C22 — Mode transitions ≤ 4 ms: pre-arm the output converter in idle, only flip its reference and enable gating.

- Task C23 — STS gating: thyristor pairs commutate at next zero-crossing detected by ZCD block; backup detection via dV/dt and V_{rms} drop within 2 ms.

5. UPS-Microgrid Controller (Block Level) — Tasks

- Task M1 — Aggregate 4–7 PCS3440 feedbacks (P, Q, Vdc, SoC, status) using PSCAD Data Labels.
- Task M2 — Block-level P_setpoint splitter using participation factors weighted by SoC and de-rating status.
- Task M3 — Block ramp shaper: a slower outer rate limit (e.g., 4 MW/min per 28 MW block, so 5 blocks total ≤ 20 MW/min) — feeds into each PCS C5.
- Task M4 — Mode arbiter: receives plant SCADA command (Double/Bypass/Backup), synchronizes all PCS state machines to flip in lock-step.
- Task M5 — STS supervisory logic: arms/disarms STS based on mode; monitors voltage/frequency window for re-closure.
- Task M6 — Energy/SoC balancer: redistributes charge among DC blocks every 1 s to keep SoC spread $< 5\%$.
- Task M7 — Q/V regulation: in Bypass mode, dispatch reactive between input and output converters running as STATCOMs.

6. Plant-Level SCADA — Tasks

- Task S1 — Five-block aggregator: sum P_block, Q_block; coordinate with utility + gas generator dispatch.
- Task S2 — Plant ramp limit at PCC: enforce 20 MW/min (TCLL \$Ramp Rate) — overrides block ramp if grid forecast tight.
- Task S3 — Intra-minute monitor: rolling 5-s window detector; if $\Delta P > 10$ MW within 5 s, command extra battery support.
- Task S4 — Gas generator dispatch logic: brings turbine on if utility import is curtailed.
- Task S5 — VPP/EMS file input (.txt) for setpoints; output logs to .out file for plotting.

7. AI Load Swing Mitigation Strategy in PSCAD

The architecture decouples the grid from the AI load using the double-conversion path; the battery + DC link absorbs the fast Δ while the input converter ramps slowly. Three layered loops implement this in the simulation:

Layer	Loop	Time Constant	What It Mitigates
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Sub-cycle (≤ 16 ms)	Output converter inner V/I loop + DC-link Cdc	0.1–2 ms	Load step transient (20→90%)
Sub-second (10 ms–1 s)	Battery current command from ΔP filter	5–50 ms	AI training pulsation, checkpoint dips
Slow (1 s – minutes)	Input ramp shaper + plant ramp limit	1–60 s	TCLL ramp / intra-minute rule at PCC

7.1 Build Tasks Specific to Mitigation

- Task A1 — Add a 1st-order LPF ($\tau = 10$ s) on the input-side P command; cutoff drives the ‘smoothed grid power’.
- Task A2 — Compute $\Delta P_{\text{fast}} = P_{\text{load}} - P_{\text{grid_smooth}}$; route to battery I^* with sign convention discharge=+.
- Task A3 — Add a 2nd faster LPF ($\tau = 1$ s) feeding super-cap-equivalent fast channel of the battery if hybrid storage modeled.
- Task A4 — Implement headroom monitor: if battery SoC $< 20\%$ or $> 80\%$, gradually relax LPF τ to let the grid see more swing (prevents battery walk-off).
- Task A5 — Add an intra-minute slew limiter on the input P command: limit Δ to 10 MW per 5-s rolling window using a moving-window max/min block.
- Task A6 — Add DC link voltage stabilizer: feed-forward of $(P_{\text{in}} - P_{\text{out}})$ into Cdc model so Vdc never droops $> 5\%$ during worst-case swing.
- Task A7 — Coordination with gas generator: if predicted $\Delta P > 50$ MW for > 30 s, send pre-start command to turbine in PLANT_SCADA.

8. Simulation Scenarios and Acceptance Criteria

#	Scenario	Trigger	Pass Criteria
S1	20→90% step in Double-Conversion	Load step at $t=1$ s	$\Delta V \leq 5\%$, recover < 16 ms
S2	90→20% step in Double-Conversion	Load drop at $t=2$ s	$\Delta V \leq 5\%$, no DC OV
S3	DC→Bypass planned transition	SCADA cmd	Transition < 4 ms, no V dip
S4	Bypass→Backup (utility loss)	Open 34.5 kV breaker	Output V uninterrupted, < 4 ms
S5	AI training pulsation 1 Hz, ± 25 MW	Profile generator	Grid P ripple < 5 MW p-p

S6	TCLL ramp test	20%→100% load over 60 s	Grid ramp ≤ 20 MW/min
S7	Intra-minute test	± 10 MW per 5 s pattern	Grid $\Delta \leq 10$ MW/5 s window
S8	Checkpoint dip	100→10% for 3 s, return	Battery absorbs excess, Vdc $\pm 5\%$
S9	Block-level fault	1 of 5 blocks trips	Remaining 4 ramp up within plant limits
S10	Black start	Battery only	Output V built up in < 200 ms

9. PCS3440 Tuning Starting Values

Loop	Kp	Ki	Bandwidth	Notes
Input dq current	0.6	300	800 Hz	Cross-decoupled, Vd/Vq FF
Input Vdc	2.0	120	60 Hz	Anti-windup ± 1 pu
Input P (ramp shaper output)	0.4	20	5 Hz	Slow on purpose
Output dq current	0.8	500	1.8 kHz	Highest bandwidth
Output Vd/Vq	1.5	400	300 Hz	Load FF mandatory
SRF-PLL	180	3200	30 Hz	SOGI preferred at weak grid
Battery I	0.5	40	10 Hz	Slope-limited per A8
Plant ramp at PCC	—	—	0.02 Hz	Rate limiter, not PI